

Research Profile

PhD candidate in reinforcement learning and optimization, graduating May 2027 and available for full-time work in Summer 2027. Develops new formalisms, algorithms, and finite-sample theory for distributional, multi-objective, risk-sensitive, and average-reward RL, as well as optimization methods for LLM post-training. Research combines mathematical analysis with implementation and empirical evaluation in tabular and neural environments.

Education

Purdue University , West Lafayette, IN	August 2022–May 2027 (expected)
PhD Candidate, Electrical and Computer Engineering	Advisor: Prof. Abolfazl Hashemi
Thesis: <i>Uncertainty, Risk, Robustness and Multi-Objective Reasoning in Sequential Decision-Making</i>	
MS, Electrical and Computer Engineering	Awarded August 2026
Boğaziçi University , Istanbul, Turkey	September 2018–June 2022
BSc, Computer Engineering and BSc, Mathematics (double major)	High honors in both majors

Research Experience

Purdue University	West Lafayette, IN
Graduate Research Assistant, MINDS Research Group	August 2022–Present

- Developed **LOO-ROLL** for LLM post-training, with finite-rank theory for EGGROLL’s update field and estimator variance. At matched training time, LOO-ROLL improved Qwen3-14B mean accuracy over EGGROLL by 7.85 percentage points on GSM8K and 8.08 percentage points on MATH-500.
- Introduced **joint MDPs**, a formalism for coupled-dynamics environments that preserves cross-action counterfactual dependence. Developed dynamic-programming and incremental algorithms with convergence guarantees and evaluated them in tabular and neural environments.
- Developed finite-iteration and structural theory for categorical distributional RL, covering asynchronous TD under i.i.d. and Markovian sampling and quotient-categorical representations for average-reward problems.
- Established convergence and oracle-complexity results for stochastic approximation and optimization, including non-expansive fixed-point iteration with Markovian data and non-convex optimization under unbounded variance.

Dolby Laboratories	San Francisco, CA
PhD Research Intern	May 2025–August 2025

- Conceived and built **ORDO**, a first-order policy-optimization method comparing multivariate return distributions through stochastic dominance, during a three-month project on experience delivery and media personalization.
- Continued the work after the internship by establishing approximate non-dominatedness guarantees and developing stochastic estimators. Published the resulting research at UAI 2026.

Additional Experience

Nokia	Istanbul, Turkey
Data Analytics and Automation Intern	July–September 2021

- Developed customer network-data processing and visualization tools using the Elasticsearch-Logstash-Grafana stack.

Scientific and Technological Research Council of Turkey (TÜBİTAK)	Istanbul, Turkey
STAR Research Intern	February–August 2021

- Studied deep-learning methods for single-image super-resolution.

Peer-Reviewed Publications

- Kaya, E.C.**, Pourghani, A., Gupta, V., Hashemi, A. “Quotient-Categorical Representations for Bellman-Compatible Average-Reward Distributional Reinforcement Learning.” **NeurIPS**, 2026. Preprint: [arXiv:2605.11289](https://arxiv.org/abs/2605.11289).
- Kaya, E.C.**, Ghasemi, M., Hashemi, A. “Joint MDPs and Reinforcement Learning in Coupled-Dynamics Environments.” **Uncertainty in Artificial Intelligence (UAI, Oral)**, 2026. Oral presentation, top 2.2% of 1,087 submissions.
- Kaya, E.C.**, Kaheman, K., Cloud, J.M., Hashemi, A. “Stochastic Dominance Driven First-Order Policy Optimization for Multi-Objective Reinforcement Learning.” **UAI**, 2026.
- Kaya, E.C.**, Hibbard, M., Tanaka, T., Topcu, U., Hashemi, A. “Randomized Greedy Methods for Weak Submodular Sensor Selection with Robustness Considerations.” **Automatica**, 171, 2025.
- Kaya, E.C.**, Hashemi, A. “Localized Distributional Robustness in Submodular Multi-Task Subset Selection.” **IEEE Transactions on Signal Processing**, 2024.

6. Jiménez, A.C.C., **Kaya, E.C.**, Ye, L., Hashemi, A. “Equitable Client Selection in Federated Learning via Truncated Submodular Maximization.” **IEEE Conference on Decision and Control (CDC)**, 2024.
7. **Kaya, E.C.**, Sahin, M.B., Hashemi, A. “Communication-Efficient Zeroth-Order Distributed Online Optimization: Algorithm, Theory, and Applications.” **IEEE Access**, 2023.
8. **Kaya, E.C.**, Hashemi, A. “Relative Entropy Regularization for Robust Submodular Multi-Task Subset Selection.” **Allerton Conference**, 2023.
9. **Kaya, E.C.**, Sahin, M.B., Hashemi, A. “Communication-Constrained Exchange of Zeroth-Order Information with Application to Collaborative Target Tracking.” **ICASSP**, 2023.
10. Jiménez, A.C.C., **Kaya, E.C.**, Hashemi, A. “High Probability Guarantees for Submodular Maximization via Boosted Stochastic Greedy.” **Asilomar Conference**, 2023.

Workshop Publications

1. **Kaya, E.C.**, Ghasemi, M., Hashemi, A. “Beyond Marginals: Capturing Correlated Returns through Joint Distributional Reinforcement Learning.” **NeurIPS Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET)**, 2025.

Manuscripts and Preprints

1. **Kaya, E.C.**, Hashemi, A. “EGGROLL, Unrolled: Understanding and Improving Low-Rank Evolution Strategies at Scale.” **ICLR 2027 (under review)**. Preprint available: [arXiv:2609.10980](https://arxiv.org/abs/2609.10980).
2. Fazla, A., Upadhyay, A., **Kaya, E.C.**, Sahin, M.B., Hashemi, A. “Why Does Adaptive Batching Help LLM Pretraining? A Perspective from Unbounded Variance.” **ICLR 2027 (under review)**.
3. Fazla, A. *, **Kaya, E.C.** *, Upadhyay, A. *, Hashemi, A. “Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance.” **ICLR 2027 (under review)**. *Equal contribution. Preprint available: [arXiv:2604.16620](https://arxiv.org/abs/2604.16620).
4. **Kaya, E.C.**, Pourghani, A., Ghasemi, M., Gupta, V., Hashemi, A. “Learning to Control Coupled-Dynamics Environments with Joint Markov Decision Processes.” **TMLR (under review)**. Preprint available: [arXiv:2608.22765](https://arxiv.org/abs/2608.22765).
5. **Kaya, E.C.**, Fazla, A., Sahin, M.B., Hashemi, A. “A Banach-Space Theory of Markovian Halpern Iteration for Non-Expansive Maps.” **ICLR 2027 (under review)**. Preprint available: [arXiv:2608.15966](https://arxiv.org/abs/2608.15966).
6. **Kaya, E.C.**, Hashemi, A. “A Finite-Iteration Theory for Asynchronous Categorical Distributional Temporal-Difference Learning.” **AISTATS 2027 (under review)**. Preprint available: [arXiv:2605.06866](https://arxiv.org/abs/2605.06866).

Teaching Experience

Purdue University

Graduate Teaching Assistant

West Lafayette, IN

August 2024–May 2025

- ECE 600: Random Variables and Signals. ECE 369: Discrete Mathematics for Computer Engineering.

Boğaziçi University

Undergraduate Teaching Assistant

Istanbul, Turkey

January–June 2022

- CMPE 160: Object-Oriented Programming.

Technical Skills

Research: reinforcement learning, distributional RL, multi-objective and risk-sensitive RL, LLM post-training, evolution strategies, stochastic approximation, optimization, probabilistic modeling, Monte Carlo methods, submodular optimization

Programming and tools: Python, PyTorch, JAX, vLLM, Ray, MATLAB, Git, Docker

Honors and Service

Bilsland Dissertation Fellowship, Purdue University. Competitive final-year doctoral fellowship.

Cahit Arf Grant, Republic of Turkey. National scholarship awarded based on top-100 placement in the university entrance examination.

Reviewer. AISTATS 2024, AAAI 2025, UAI 2025–2026, NeurIPS 2026, ICLR 2027, and NeurIPS ARLET Workshop 2025.

Languages: Turkish (native), English (proficient), French (advanced).